

2.12. Source Sheets and Panels (Rough Draft)

We have now seen how two or more elementary flows can be combined to produce a more complex flow over some body of interest. By adding a source flow to a uniform flow we produced flow over a semi-infinite body of some specific shape. Combining uniform flow with both a source and a sink produced flow over a Rankine oval. Flow over a circular cylinder was produced from uniform flow combined with doublet flow. In the development of each of these compound potential flows, we started with an apparently arbitrary combination of elementary flows and discovered that the resulting streamlines were consistent with flow over a body of some special shape. This indirect method of starting with some combination of elementary flows and examining the resulting body shape is obviously not practical for solving the more general problem of flow over a body of known but arbitrary shape. There is no way of knowing in advance the correct combination of elementary flows that are needed to synthesize the flow over a body of any specific shape. For this more general problem we would favor a direct method which allows us to first specify the shape of the body and then solve for the distribution of elementary flows that will exactly produce flow over that given body. Such a method requires the consideration of some additional elementary potential flows.

While it is possible to synthesize flow over some bodies of very specific shape by combining a finite number of sources and sinks with a uniform flow, to synthesize flow over a body of arbitrary shape, we require greater flexibility. In this section we shall examine yet another elementary 2-dimensional potential flow that will help provide this increased flexibility. This new potential flow is formed by combining an infinite number of infinitesimally weak line-sources in a side-by-side fashion as shown in Fig. 2.12.1. In this figure we are looking at an edge view of the sheet with the line-sources all perpendicular to the page. Let s be the distance measured along this source sheet as shown in Fig 2.12.1 and let $\lambda(s)$ be the source strength per unit length measured along s . The total strength associated with a differential segment of the source sheet located at s and having length ds is then $\lambda(s)ds$. Thus, the differential velocity induced by ds at an arbitrary point in space (x,y) , located a distance r from ds , is given by

$$d\mathbf{V} = \frac{\lambda(s)}{2\pi r} \mathbf{i}_r ds \quad (2.12.1)$$

where $\mathbf{i}_r$ is the unit vector in the direction from ds to the point (x,y) . The total velocity induced at the point (x,y) by the entire source sheet, extending from $s = 0$ to $s = l$, is then

$$\mathbf{V}(x,y) = \frac{1}{2\pi} \int_{s=0}^l \frac{\lambda(s)}{r} \mathbf{i}_r ds \quad (2.12.2)$$

While a source sheet with a completely arbitrary curved shape, such as that shown in Fig. 2.12.1, is required to exactly synthesize flow over a body of arbitrary shape, such a sheet can be closely approximated using a series of planar segments. A planar source sheet is usually referred to as a source panel. As will be seen in the next chapter, approximating a source sheet with a series of source panels significantly simplifies the solution process.

Consider a differential segment of a 2-dimensional source panel that lies on the x -axis at the location $x = x_o$. The velocity induced by this differential source panel, at an arbitrary point with coordinates (x,y) , is given by

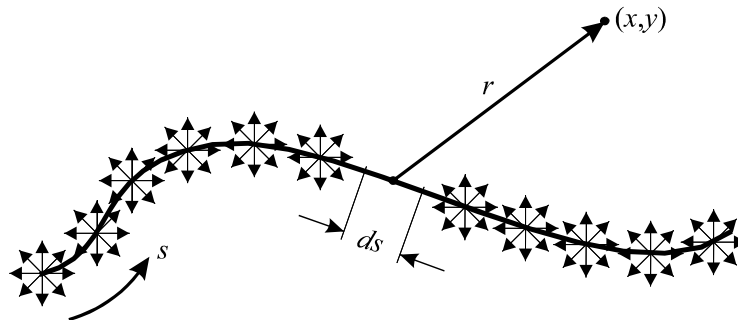


Figure 2.12.1. Edge view of a 2-dimensional source sheet.

$$dV_r = \frac{\lambda(x_o)}{2\pi r} dx_o \quad (2.12.3)$$

$$dV_\theta = 0 \quad (2.12.4)$$

where $\lambda(x_o)$ is the source strength per unit length at x_o and the remaining notation is defined in Fig. 2.12.2. The x - and y -components of this velocity are

$$dV_x = dV_r \cos \theta = \frac{\lambda(x_o)}{2\pi r} \cos \theta dx_o \quad (2.12.5)$$

$$dV_y = dV_r \sin \theta = \frac{\lambda(x_o)}{2\pi r} \sin \theta dx_o \quad (2.12.6)$$

We can also write

$$r = \sqrt{(x-x_o)^2 + y^2} \quad (2.12.7)$$

$$\cos \theta = \frac{x-x_o}{\sqrt{(x-x_o)^2 + y^2}} \quad (2.12.8)$$

$$\sin \theta = \frac{y}{\sqrt{(x-x_o)^2 + y^2}} \quad (2.12.9)$$

Using Eqs. (2.12.7), (2.12.8), and (2.12.9) in Eqs. (2.12.5) and (2.12.6), we have

$$dV_x = \frac{(x-x_o)\lambda(x_o)}{2\pi[(x-x_o)^2 + y^2]} dx_o \quad (2.12.10)$$

$$dV_y = \frac{y\lambda(x_o)}{2\pi[(x-x_o)^2 + y^2]} dx_o \quad (2.12.11)$$

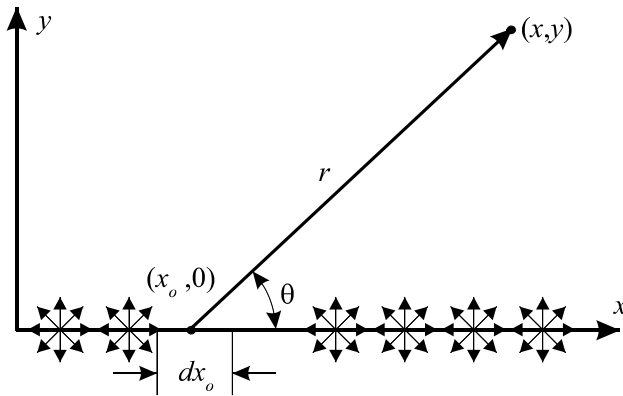


Figure 2.12.2. Edge view of a 2-dimensional source panel on the x -axis.

For a finite 2-dimensional source panel that extends from $x_o = 0$ to $x_o = l$, we can express the velocity induced at the point (x, y) by the entire source panel as,

$$V_x(x, y) = \frac{1}{2\pi} \int_{x_o=0}^l \frac{(x-x_o)\lambda(x_o)}{(x-x_o)^2 + y^2} dx_o \quad (2.12.12)$$

$$V_y(x, y) = \frac{y}{2\pi} \int_{x_o=0}^l \frac{\lambda(x_o)}{(x-x_o)^2 + y^2} dx_o \quad (2.12.13)$$

If the strength distribution is known, the right-hand side of Eqs. (2.12.12) and (2.12.13) can be integrated to obtain the velocity field. For numerical solutions, a linear strength distribution is often used. For this strength distribution we have

$$\lambda(x_o) = \lambda_1 - \frac{x_o}{l}(\lambda_1 - \lambda_2) \quad (2.12.14)$$

$$V_x(x, y) = \frac{1}{2\pi} \int_{x_o=0}^l \frac{(x-x_o)\left[\lambda_1 - \frac{x_o}{l}(\lambda_1 - \lambda_2)\right]}{(x-x_o)^2 + y^2} dx_o \quad (2.12.15)$$

$$V_y(x, y) = \frac{y}{2\pi} \int_{x_o=0}^l \frac{\lambda_1 - \frac{x_o}{l}(\lambda_1 - \lambda_2)}{(x-x_o)^2 + y^2} dx_o \quad (2.12.16)$$

where λ_1 and λ_2 are the strengths at $x_o = 0$ and $x_o = l$ respectively. Evaluating the integrals on the right-hand side of Eqs. (2.12.15) and (2.12.16) we obtain

$$\begin{aligned} V_x(x, y) = & -\frac{y(\lambda_1 - \lambda_2)}{2\pi l} \left[\tan^{-1}\left(\frac{x}{y}\right) - \tan^{-1}\left(\frac{x-l}{y}\right) \right] \\ & + \frac{\lambda_1 l - x(\lambda_1 - \lambda_2)}{4\pi l} \ln \left[\frac{x^2 + y^2}{(x-l)^2 + y^2} \right] + \frac{\lambda_1 - \lambda_2}{2\pi} \end{aligned} \quad (2.12.17)$$

$$\begin{aligned} V_y(x, y) = & \frac{\lambda_1 l - x(\lambda_1 - \lambda_2)}{2\pi l} \left[\tan^{-1}\left(\frac{x}{y}\right) - \tan^{-1}\left(\frac{x-l}{y}\right) \right] \\ & + \frac{y(\lambda_1 - \lambda_2)}{4\pi l} \ln \left[\frac{x^2 + y^2}{(x-l)^2 + y^2} \right] \end{aligned} \quad (2.12.18)$$

The most time consuming computations in Eqs. (2.12.17) and (2.12.18) are the logarithm function and the two arctangent functions. We can simplify the two arctangent functions by using the trigonometric relation,

$$\tan(\Phi_1 - \Phi_2) = \frac{\tan(\Phi_1) - \tan(\Phi_2)}{1 + \tan(\Phi_1)\tan(\Phi_2)} \quad (2.12.19)$$

If we let $\Phi_1 = \tan^{-1}(x/y)$ and $\Phi_2 = \tan^{-1}[(x-l)/y]$, equation (2.12.19) gives

$$\tan^{-1}\left(\frac{x}{y}\right) - \tan^{-1}\left(\frac{x-l}{y}\right) = \tan^{-1} \left[\frac{\left(\frac{x}{y}\right) - \left(\frac{x-l}{y}\right)}{1 + \left(\frac{x}{y}\right)\left(\frac{x-l}{y}\right)} \right] = \tan^{-1} \left(\frac{yl}{y^2 + x^2 - xl} \right) \quad (2.12.20)$$

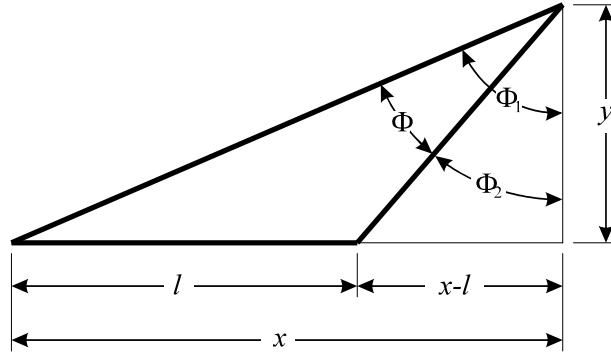


Figure 2.12.3. Geometry for a finite 2-dimensional source panel.

Using Eq. (2.12.20) in Eqs. (2.12.17) and (2.12.18), the velocity induced by a **2-dimensional source panel with linear strength distribution along the x-axis**, having length l with $\lambda(0) = \lambda_1$ and $\lambda(l) = \lambda_2$, is found to be

$$\begin{Bmatrix} V_x \\ V_y \end{Bmatrix} = \frac{1}{2\pi l} \begin{bmatrix} [-y\Phi + (l-x)\Psi + l] & (y\Phi + x\Psi - l) \\ [(l-x)\Phi + y\Psi] & (x\Phi - y\Psi) \end{bmatrix} \begin{Bmatrix} \lambda_1 \\ \lambda_2 \end{Bmatrix} \quad (2.12.21)$$

where

$$\Phi = \text{atan2}(y, y^2 + x^2 - xl) \quad (2.12.22)$$

and

$$\Psi = \frac{1}{2} \ln \left[\frac{x^2 + y^2}{(x-l)^2 + y^2} \right] \quad (2.12.23)$$

As can be seen in Fig. 2.12.3, the angle Φ can take any value between $-\pi$ and π . Note that when the point (x, y) is either left or right of the panel ($x < 0$ or $x > l$) the angle Φ is positive when y is above the axis, negative when y is below the axis, and approaches zero as y approaches zero from either side. However, when the point (x, y) is directly above the panel ($0 < x < l$ and $y > 0$) and y approaches zero from above, the angle Φ approaches π . On the other hand, when the point (x, y) is directly below the panel ($0 < x < l$ and $y < 0$) and y approaches zero from below, the angle Φ approaches $-\pi$. Thus, we see that there is a step discontinuity of 2π in the angle Φ as we cross the panel itself,

$$\Phi(x, \pm 0) = \pm \pi, \quad 0 < x < l \quad (2.12.24)$$

where $y = +0$ indicates a point just above the surface and $y = -0$ indicates a point just below the surface.

Single-argument arctangent functions like the fortran or C intrinsic function “atan” will always return values between $-\pi/2$ and $\pi/2$. Such functions cannot be used to compute the angle Φ from Eq. (2.12.22) for all values of x and y . To obtain the correct result over the full range of x and y , the angle Φ must always be computed using a two-argument arctangent function, such as the fortran or C function “atan2.”

Using Eq. (2.12.24) in Eqs. (2.12.17) and (2.12.18), we see that the velocity along the upper and lower surfaces of the source panel ($0 < x < l$ and $y > \pm 0$) is given by

$$V_x(x, \pm 0) = \frac{\lambda_1 l - x(\lambda_1 - \lambda_2)}{4\pi l} \ln \left[\frac{x^2}{(x-l)^2} \right] + \frac{\lambda_1 - \lambda_2}{2\pi} \quad (2.12.25)$$

$$V_y(x, \pm 0) = \pm \frac{\lambda_1 l - x(\lambda_1 - \lambda_2)}{2l} \quad (2.12.26)$$

So we see that the tangential velocity is continuous across the source panel while there is a step change in the normal velocity. The change in normal velocity across the panel is

$$V_y(x,+0) - V_y(x,-0) = \lambda_1 - \frac{x}{l}(\lambda_1 - \lambda_2) \quad (2.12.27)$$

or in view of Eq. (2.12.14)

$$V_y(x,+0) - V_y(x,-0) = \lambda(x) \quad (2.12.28)$$

Equation (2.12.28) points out a very important relation that always holds true for any source sheet, planar or not. That is, at any point along a source sheet, the local jump in normal velocity across the sheet is always equal to the local sheet strength. There is no tangential velocity jump across a source sheet.